QP CODE: 25900020

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBOT100 - GARDENING AND LANDSCAPING
(2024 ADMISSION ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall
outcome attainability

Duration : 1Hour | Max. Marks : 35

Part A -
Short Answer- Answer any 5 questions - Each question carries 1 marks

1	Mention any one importance of landscaping.		[U] / [CO1]
		Landscaping is important for its economic, environmental, and lifestyle benefits. It enhances property values, improves air and water quality, reduces pollution, and promotes recreation and mental well-being. **(any one importance)**	
2	In what way does organic mulch improve soil health?		[U] / [CO1, CO2]
		Organic mulch improves soil health by adding essential nutrients as it decomposes, conserving moisture, regulating soil temperature, and enhancing soil structure. It also helps suppress weed growth and encourages beneficial soil organisms.	
3	Name a bulbous crop used in landscaping		[K] / [CO2]
		Tulip, daffodils, lilies, gladioli etc (any one).	
4	What is layering ?		[U] / [CO4]
		Layering is a plant propagation technique where new roots develop on a stem while it's still attached to the parent plant. Once the roots form, the stem is separated and can grow as a new, independent plant.	
5	Name the light colours used to create calming effect in landscape.		[K] / [CO2, CO5]
		To create a calming effect in a landscape - using light shades of blue, green, lavender, and shades of white can used. These colors are often associated with peace, tranquility, and relaxation.	
6	What is the primary purpose of site analysis in landscape design?		[U] / [CO2]
		Site analysis is crucial in landscape architecture because it provides a foundation for understanding and responding to a site's physical, environmental, social, and historical context.	
7	What are the elements in landscape design?		[K] / [CO2, CO5
		The key elements of landscape design include line, form, texture, color, and scale	
8		Interpret the role of garden seats in enhancing visitor experience.	[U] / [CO2

		Garden seats enhance the visitor experience by offering comfort, promoting relaxation, and fostering social interaction.	
			(1x5 = 5)

Part B
Short Essay - Answer any 4 questions - Each question carries 5 marks

9	Discuss the role of Aglaonema plants in interior design.		[U] / [CO1, CO2]
		Aglaonema, commonly known as Chinese Evergreen, is a genus of flowering plants in the Araceae family (Arums) It thrives best in indirect bright light, they also adapt to low light situations making it easy to grow indoors **Aesthetic and Functional Benefits**: Greenery and Nature: Aglaonema's vibrant foliage provides a refreshing pop of color and nature, making spaces feel more inviting and alive. Low-Maintenance: They are relatively easy to care for, making them ideal for homes or offices where owners may not have a lot of time for plant care. Air Purification: Aglaonema are known to remove common indoor pollutants like formaldehyde, benzene, and other VOCs, contributing to a healthier indoor environment. Versatile: Their diverse foliage and varied color options allow for incorporation into a wide range of design styles, from modern to traditional. Focal Point: Aglaonema can be used as a statement piece, adding visual interest to a room. **Symbolic and Cultural Significance:** Good Luck: In many cultures, including those in Asia, Aglaonema is considered a lucky plant, symbolizing prosperity, health, and positive energy.	
10	Name two styles of ornamental gardens and briefly describe their key features.		[K] / [CO1, CO2]
		Two popular styles of ornamental gardens are formal and informal gardens. Formal gardens are characterized by symmetry and geometric shapes, while informal gardens embrace asymmetry and naturalistic designs. Formal Gardens: Key futures **Symmetry**: Formal gardens are meticulously planned with symmetrical layouts, where mirrored plantings and geometric flower beds are common. **Geometric Shapes**: Elements like flower beds, pathways, and fountains are arranged in precise, geometric patterns. **Trimmed Hedges**: Hedges are often used to create structured boundaries and further enhance the symmetrical look. **Central Axis**: Many formal gardens feature a central axis, with symmetrical features	

		arranged on either side. Examples: Mughal gardens, Italian gardens, and French gardens are well-known examples of formal garden styles. **Informal Gardens**: **Asymmetry**: Informal gardens avoid strict symmetry and instead embrace a more natural, irregular design. **Naturalistic Planting**: Plants are arranged in a more free-flowing manner, mimicking natural groupings. **Curved Paths**: Paths are often curved and winding, rather than straight lines. **Unstructured Elements**: Features like ponds, rocks, and trees are placed in a less rigid and more naturalistic way. Examples: English gardens, Japanese gardens, and wild gardens often incorporate informal design principles.	
11	What are the applications of vegetative propagation ?		[U] / [CO4]
		Applications of vegetative propagation: **Horticulture:** ➢ Commercial production: Allows for the mass production of plants with specific characteristics, ensuring consistency in quality and taste, e.g. apples, bananas, etc ➢ Propagation of seedless plants: Enables the reproduction of plants like bananas and oranges that have lost the ability to produce viable seeds, ➢ Maintaining genetic uniformity: Ensures that all offspring are identical to the parent plant, preserving desirable traits. **Forestry:** ➢ Establishment of clonal seed orchards: Used to propagate trees with desirable traits for seed production, . ➢ Establishment of clonal banks: Used to store and propagate genetically diverse tree populations for research or conservation, ➢ Propagation of special breeding material: Helps preserve exceptional hybrids or sterile hybrids that would be lost through sexual reproduction, **Conservation:** ➢ Preservation of rare and endangered species: Facilitates the propagation of plants that are difficult to reproduce through seed, ➢ Introduces plants to new areas: Allows for the establishment of plants in areas where seed germination might not be successful, **Other applications:** ➢ Quickly producing plants: Offers a faster and easier method of reproduction compared to seed propagation, especially for plants that are slow to mature or have low seed viability,. ➢ Avoiding costly and complicated processes: Allows plants to bypass the intermediate seedling stage and reach maturity more quickly,	

		(Any five relevant applications – Full credit)	
12	Comment on the uses of budding and grafting knife.		[U] / [CO2]
		Uses of budding and grafting knife 1. Used to make precise cuts on plant stems and branches during grafting and budding procedures. 2. Lifting of bark during budding operation 3. Cutting of thin stems, roots and dried inflorescence during pruning operation 4. Extraction bud from budwood 5. To prune water sprouts 6. Pruning of dried and diseased thin stem and other plant parts 7. To perform notching in budded stumps (any five uses)	
13	Write a brief outline of a rockery		[U] / [CO2, CO3]
		Rockery, also known as a rock garden or alpine garden, is a garden component featuring rocks and stones, typically with alpine plants that thrive in well-draining, sunny conditions. **Purpose**: Rockeries aim to replicate the harsh, rocky environments where alpine plants naturally grow, providing good drainage and ample sunlight. **Plants**: Alpine plants, small shrubs, and perennials that can tolerate poor soil and dry conditions are commonly used in rockeries. **Design**: Rocks and stones are arranged to create a visually appealing landscape, with plants interspersed between them. **Maintenance**: Once established, rockeries require relatively little maintenance, as they are adapted to withstand harsh conditions. **Benefits**: Rockeries add visual interest and structure to a garden, providing diverse habitats and creating a peaceful space	
14	How does a rain shelter differ from a polyhouse in terms of structure and function?		[K] / [CO3]
		Rain shelters are simpler structures, typically with a roof made of a waterproof material like plastic sheeting or netting, often with open sides for natural ventilation. Polyhouses, , are more complex, featuring a fully enclosed structure, often with sidewalls and a roof made of polythene or other materials, and may include climate control systems. **Key Differences in Structure**: **Enclosure** : Rain shelters are usually not fully enclosed, often having open sides or only a roof covering, while polyhouses are typically fully enclosed. **Materials** : Rain shelters may use simpler materials like plastic sheeting or netting, while polyhouses often use polythene or other materials with more robust frames. **Ventilation**: Rain shelters rely on natural ventilation, with open sides, while polyhouses may have mechanical ventilation systems or rely on natural air convection. **Climate Control**: Rain shelters offer limited or no climate control, while polyhouses may have systems for controlling temperature, humidity, and other environmental factors. **Purpose**: Rain shelters are primarily used for protecting crops from heavy rainfall and other	

		weather conditions, while polyhouses are designed for more controlled growing environments and a wider range of crops.	
			(5 × 4 = 20)

Part C **Essay Answer any 1 questions - Each question carries 10 marks**			
15	Explain Budding techniques. Add comment on its advantages and disadvantages.		[U] / [CO4]
		Budding is the art of inserting a bud on the root stock plant in such a manner that both will unite and continue to grow as a single individual plant The shoots of mother tree from which buds are taken for use as scion material for budding are called bud wood. Bud wood should collected from selected trees and buds should be dormant (brown colored). Types of Budding: T-budding: A common method where a T-shaped cut is made in the rootstock, and a bud is inserted. Chip budding: A bud is inserted into a chip-shaped cut in the rootstock. Patch budding: A larger patch of bark containing a bud is inserted. Ring budding: A ring of bark is removed from the rootstock and replaced with a bud. **<u>T-budding - procedure</u>** 1.Scion and rootstock: Select a bud (scion) from a desired plant and a rootstock (usually of a different variety). 2.Preparing the rootstock: Make the appropriate cut (e.g., T-shaped or chip) on the rootstock stem. 3.Inserting the bud: Insert the bud into the prepared cut, ensuring close contact between the cambium layers of both scion and rootstock. 4.Binding: Securely bind the scion bud to the rootstock to facilitate healing and prevent drying out. 5.Aftercare: Once the bud has healed, the rootstock may need to be removed to allow the scion to grow. **<u>Patch Budding -Procedure:</u>** 1. Prepare the rootstock: - Select a smooth, healthy rootstock, preferably one-year-old. Clean the rootstock by removing leaves and side branches. 2. Prepare the budwood: - Choose a budstick with healthy, plump buds from a desired cultivar. Cut the budwood about 2-3 weeks before budding, leaving leaf petioles intact. 3. Create the patch: Using a double-bladed knife or a knife with a template, create a rectangular patch (approximately 1 inch square) on the budwood, including the bud. 4. Create the patch on the rootstock: Make two parallel horizontal cuts on the rootstock, followed by vertical cuts at each end, creating a patch of the same size as the one on the budwood. 5. Insert the bud patch: - Gently peel back the bark on the rootstock and insert the bud	

		patch, ensuring the cambium layers of the scion and rootstock align. 6. Secure the patch: Fold the rootstock bark flaps back to cover the bud patch and tie it securely with budding tape or rubber. 7. Aftercare: --Leave the bud exposed and allow the patch to heal for 2-3 weeks. Remove the tape after the buds have healed, and remove the stock above the bud union, encouraging growth from the scion bud. **<u>Chip budding – Procedure</u>** In one side of the rootstock, a sanding cut of 2.5 cm in length going to a depth of half of the thickness of the stock is made. A notch is given at the bottom end of the cut. A similar cut is made in the scion shoot and a bud with a large piece of wood is removed. The scion bud is exactly fitted into the cut made in the rootstock and tied. E.g. Grapes. **<u>Advantages of Budding:</u>** Rapid reproduction: Budding allows for quick generation of new individuals, which can be advantageous in stable environments or when resources are abundant. Energy efficiency: Budding doesn't require energy-intensive processes like gamete formation, making it a more efficient reproductive strategy. Suitable for stable environments: In stable environments, the lack of genetic diversity may not be a significant disadvantage as the cloned offspring are well-adapted to their environment. Faster fruit bearing in plants: Budding can be used to produce fruit more quickly than grafting in some plant species. **<u>Disadvantages of Budding:</u>** Lack of genetic diversity: Budding produces genetically identical offspring, limiting the potential for adaptation to changing environments. Vulnerability to disease: The lack of genetic diversity can make populations more susceptible to diseases that affect the parent organism. Limited adaptability: In variable environments, the lack of genetic variation can make it difficult for the cloned offspring to survive and adapt to new conditions. **(Budding – technique - explanation 6 marks, Advantages and disadvantages – 2 marks each)**
16	Write an essay on any five principles of landscape design		[U] / [CO2, CO5
		Principles of landscape Design Landscaping is making of pictures with plant material and, hence, its principles are same as those of art. • They are as follows: 1.<u>Rhythm</u> - Repetition of same object at equidistance is called rhythm. In Mogul gardens, fountains and water canals have also been extensively used to create such effect. 2. <u>Balance</u> - The balance in landscape design is visual equilibrium of different garden elements. Balance can be created in a garden either formal or informal by grouping the components, structures and plants equally on both sides of the imaginary central axis.	

		3. Accent or emphasis or Focal point - The accent or emphasis is created in the gardens to avoid the monotonous view. • This also serves as the centre of attraction. • Mostly unusual objects like tall fountain, tree, statue etc. are used to create the effect of accent or emphasis. 4. Contrast: - This principle is most useful in emphasising the best features of an object. • It can be very easily understood by following contrast colour theory. • Against green background, a fleck of scarlet colour will make a contrast and will make scarlet colour prominent. 5. Scale or Proportion - When two or more objects are put together the proportions are established. • In a landscape design, space provided for lawn, paths, herbaceous borders, shrubbery border, trees, buildings and other garden objects should be in a right proportion. • It will create harmonious effect and look better. 6. Harmony or order or Unity - It is an overall effect of various features, styles, and colour schemes of the total scene. • The degree of harmony or unity of various elements of landscape is a measure induced in us and is called as beauty. 7.Transition/Mobility - It is the gradual change. • It could be achieved through shape, color and position of material. • Gradual change in pattern, slow change in style rather than at once is preferable in transition. • Transition can be obtained by the arrangement of objects with varying textures, forms or sizes in a logical sequential order. (**any five principles with explanation – full credit**)
		(10 × 1 = 10)